

Week_5

```
pran@DESKTOP-0NIMS2G: ~/OSSP
pran@DESKTOP-0NIMS2G:~$ gedit week_5
pran@DESKTOP-0NIMS2G:~$ gcc week_5.c
pran@DESKTOP-0NIMS2G:~$ gcc week_5 -o week5
pr/bin/ld: cannot find week_5: No such file or directory
collect2: error: ld returned 1 exit status
pran@DESKTOP-0NIMS2G:~$ ./week_5
ash: ./week_5: No such file or directory
pran@DESKTOP-0NIMS2G:~$ cd OSSP
pran@DESKTOP-0NIMS2G:~/OSSP$ gedit week_5.c
pran@DESKTOP-0NIMS2G:~/OSSP$ gcc week_5 -o week5
pr/bin/ld: cannot find week_5: No such file or directory
collect2: error: ld returned 1 exit status
pran@DESKTOP-0NIMS2G:~/OSSP$ gcc week_5 -o week_5
cc: fatal error: input file 'week_5' is the same as output file
compilation terminated.
pran@DESKTOP-0NIMS2G:~/OSSP$ gcc week_5.c -o week_5
pran@DESKTOP-0NIMS2G:~/OSSP$ ./week_5
ls
include
scripts
c
week_5.c
pran@DESKTOP-0NIMS2G:~/OSSP$ gedit week_5.c
```

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <unistd.h>
4 #include <sys/wait.h>
5
6 int main() {
7     int fd[2];
8
9     if (pipe(fd) == -1) {
10         perror("pipe");
11         return 1;
12     }
13
14     pid_t pid = fork();
15
16     if (pid < 0) {
17         perror("fork");
18         return 1;
19     }
20
21     if (pid > 0) {
22         // Parent process: executes ls -l
23
24         close(fd[0]);
25         dup2(fd[1], STDOUT_FILENO);
26         close(fd[1]);
27
28         execlp("ls", "ls", "-l", NULL);
29
30         perror("ls failed");
31         exit(1);
32     }
33     else {
34         // Child process: executes grep ".c"
35
36         close(fd[1]);
37         dup2(fd[0], STDIN_FILENO);
38         close(fd[0]);
39
40         execlp("grep", "grep", ".c", NULL);
41
42         perror("grep failed");
43         exit(1);
44     }
45
46     return 0;
47 }
```